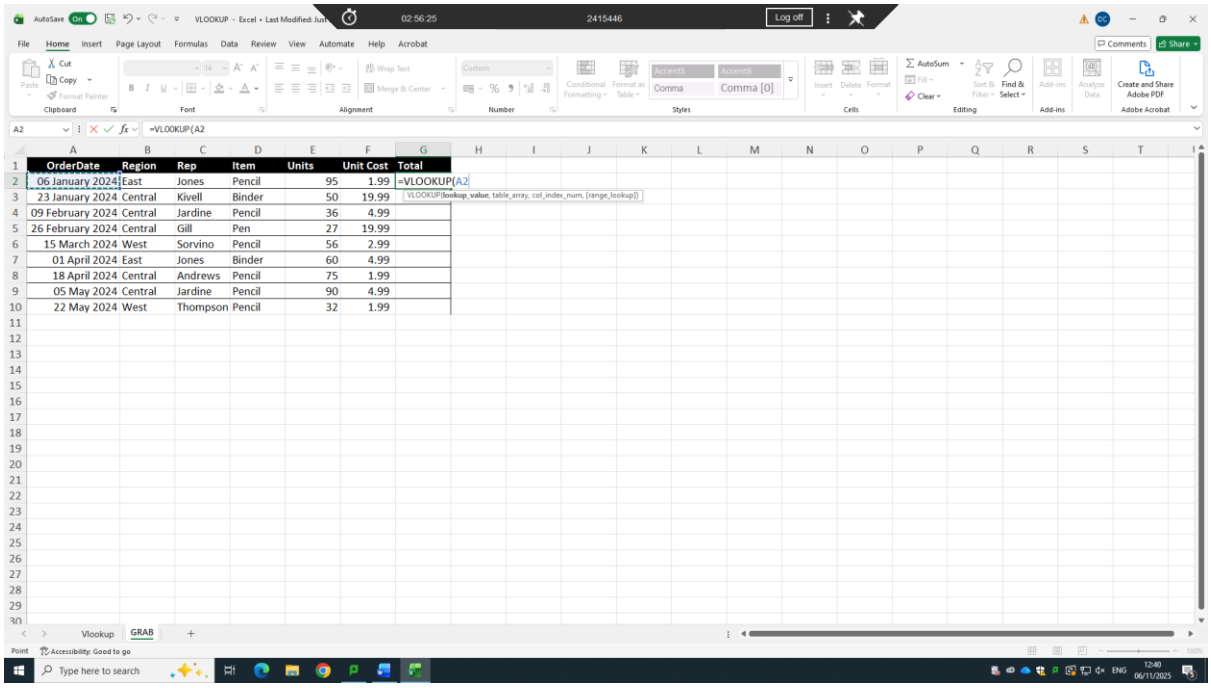
Excel Data Analysis: VLOOKUP and IF Statements
Applied Computing – Derek Agyei

VLOOKUP- VLOOKUP – Data Retrieval.

To retrieve the Total value for each order from the dataset using a lookup based on the order date, ensuring consistent and accurate data retrieval without manual entry.

Figure 1: Dataset used for VLOOKUP

This screenshot shows the main dataset containing order details such as order date, region, item, units sold, unit cost, and total value. The table serves as both the source data and the lookup range for the VLOOKUP function. The Unit Cost column was pre-existing within the dataset and was not calculated as part of this task.



OrderDate	Region	Rep	Item	Units	Unit Cost	Total
06 January 2024	East	Jones	Pencil	95	1.99	=VLOOKUP(A2
23 January 2024	Central	Kivell	Binder	50	19.99	
09 February 2024	Central	Jardine	Pencil	36	4.99	
26 February 2024	Central	Gill	Pen	27	19.99	
15 March 2024	West	Sorvino	Pencil	56	2.99	
01 April 2024	East	Jones	Binder	60	4.99	
18 April 2024	Central	Andrews	Pencil	75	1.99	
05 May 2024	Central	Jardine	Pencil	90	4.99	
22 May 2024	West	Thompson	Pencil	32	1.99	

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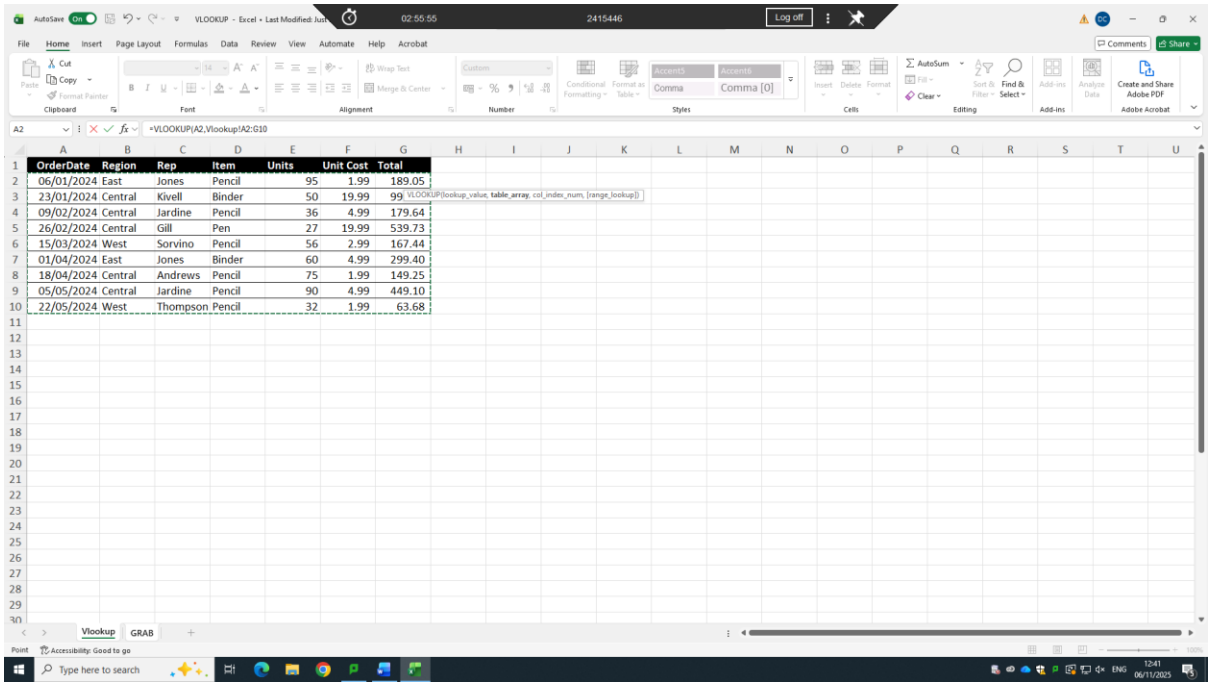

OrderDate	Region	Rep	Item	Units	Unit Cost	Total
06/01/2024	East	Jones	Pencil	95	1.99	189.05
23/01/2024	Central	Kivell	Binder	50	19.99	99
09/02/2024	Central	Jardine	Pencil	36	4.99	179.64
26/02/2024	Central	Gill	Pen	27	19.99	539.73
15/03/2024	West	Sorvino	Pencil	56	2.99	167.44
01/04/2024	East	Jones	Binder	60	4.99	299.40
18/04/2024	Central	Andrews	Pencil	75	1.99	149.25
05/05/2024	Central	Jardine	Pencil	90	4.99	449.10
22/05/2024	West	Thompson	Pencil	32	1.99	63.68

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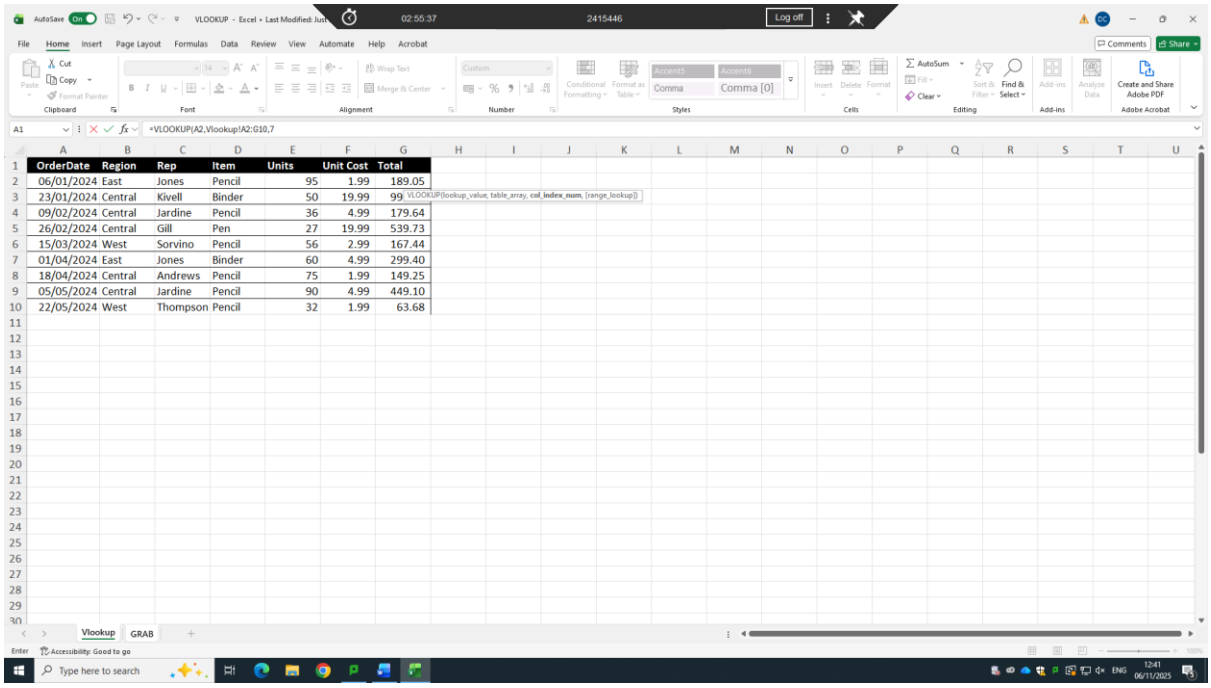

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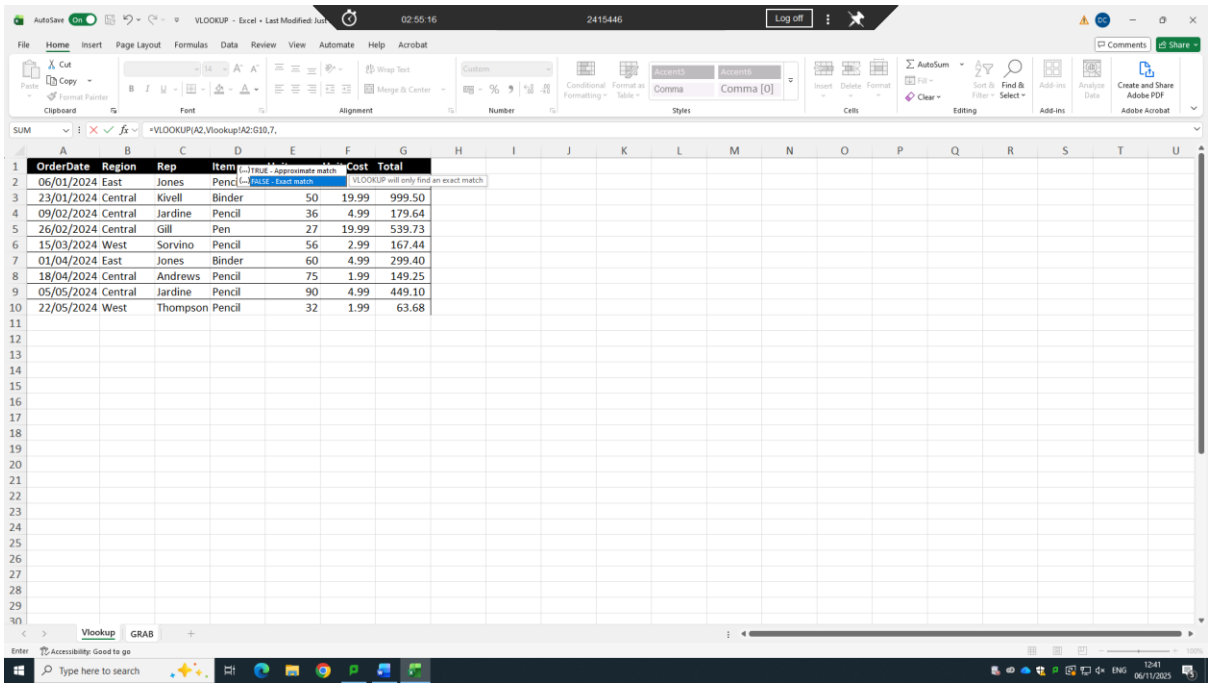
22/05/2024	West	Thompson	Pencil	32	1.99	63.68

Figure 4: The formula searches for the Order Date in cell A2 within the lookup table on the *VLOOKUP* worksheet. The column index number **7** specifies that the Total value should be returned. The **FALSE** argument ensures an exact match, preventing incorrect results.



OrderDate	Region	Rep	Item	Cost	Total
06/01/2024	East	Jones	Pencil		
23/01/2024	Central	Kivell	Binder	19.99	999.50
09/02/2024	Central	Jardine	Pencil	4.99	179.64
26/02/2024	Central	Gill	Pen	19.99	539.73
15/03/2024	West	Sorvino	Pencil	2.99	167.44
01/04/2024	East	Jones	Binder	4.99	299.40
18/04/2024	Central	Andrews	Pencil	1.99	149.25
05/05/2024	Central	Jardine	Pencil	4.99	449.10
22/05/2024	West	Thompson	Pencil	1.99	63.68

This exercise demonstrates the practical use of Excel functions to automate data retrieval, calculation, and classification.

By combining VLOOKUP, IF statements, autofill, and conditional formatting, the dataset becomes easier to analyse, reduces human error, and supports faster decision-making. These techniques are directly applicable to office administration, data handling, and entry-level IT roles.

Figure 5 – VLOOKUP Formula Before Autofill

Shows the VLOOKUP formula applied to the first row of the *Total* column (cell G2) before autofill is used. At this stage, the formula is correctly returning the total value for the first order only.

The formula references the lookup table on the *Vlookup* worksheet and uses an exact match to ensure accuracy. Applying the formula initially to a single row allows verification that the correct value is returned before extending the formula to the remaining records. This step confirms the correctness of the formula prior to replication across the dataset.

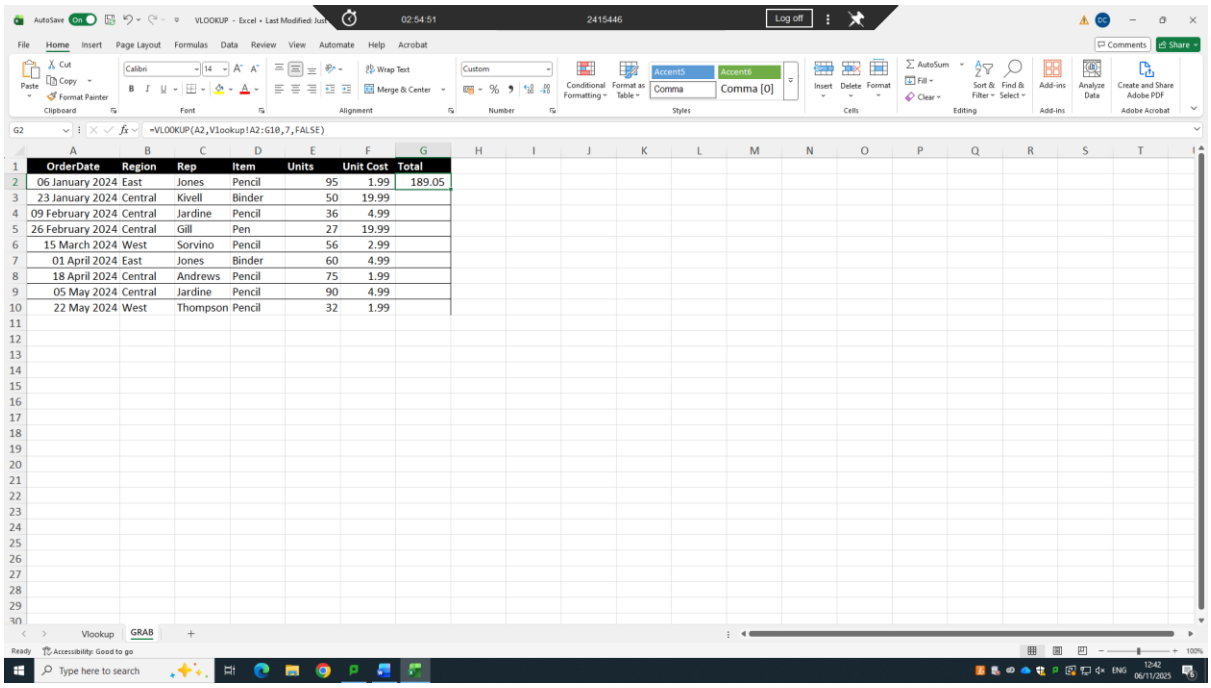
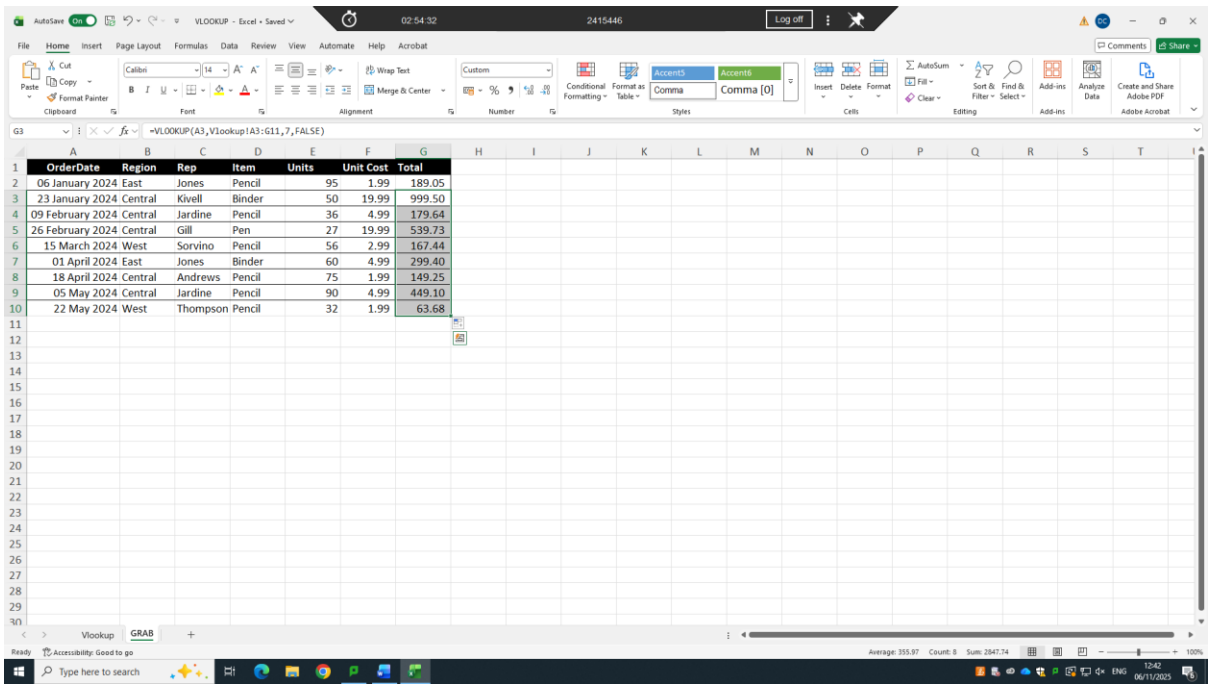


Figure 6: Applying VLOOKUP using Autofill

After entering the VLOOKUP formula in the first row of the Total column, Excel's autofill feature was used to apply the same formula to all remaining rows in the dataset. This allows the total value to be calculated automatically for each order without manually re-entering the formula.



OrderDate	Region	Rep	Item	Units	Unit Cost	Total
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IF **Figure 1: IF Formula Entry**

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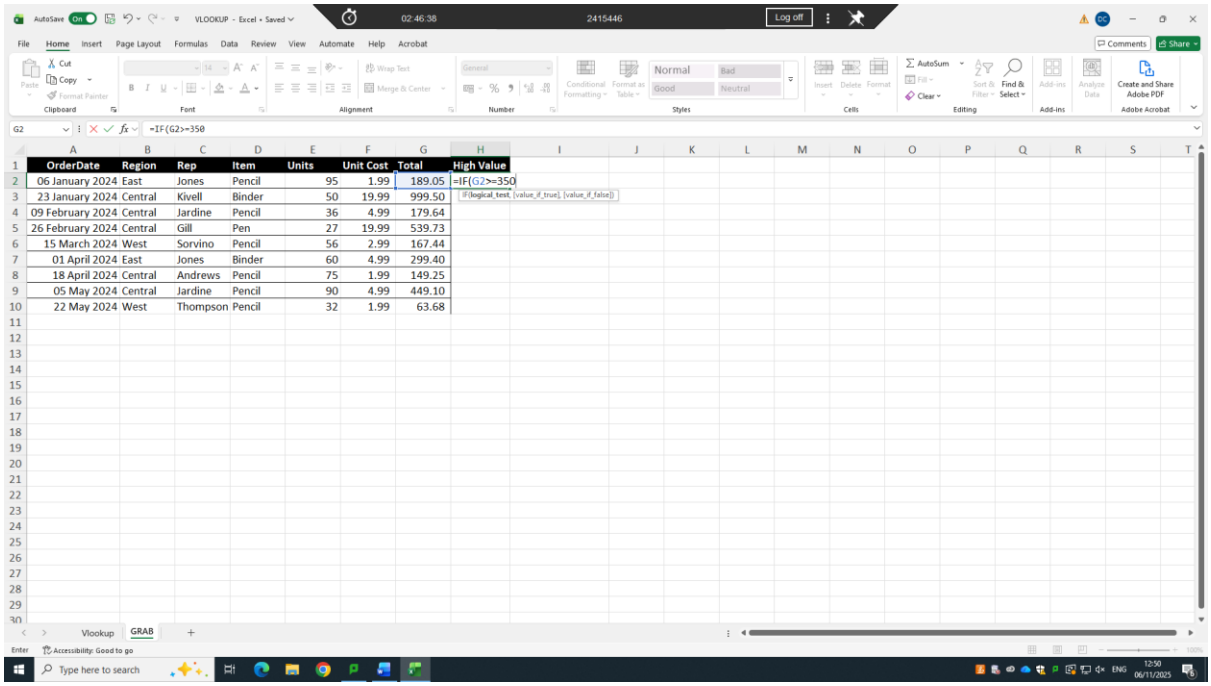

OrderDate	Region	Rep	Item	Units	Unit Cost	Total	High Value
06 January 2024	East	Jones	Pencil	95	1.99	189.05	=IF(G2>=350
23 January 2024	Central	Kivell	Binder	50	19.99	999.50	
09 February 2024	Central	Jardine	Pencil	36	4.99	179.64	
26 February 2024	Central	Gill	Pen	27	19.99	539.73	
15 March 2024	West	Sorvino	Pencil	56	2.99	167.44	
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05 May 2024	Central	Jardine	Pencil	90	4.99	449.10	
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An IF statement was implemented to classify orders based on their total value.

If the Total value in column G is greater than or equal to 350, the order is labelled as **"High"**; otherwise, it is labelled **"Normal"**. This introduces basic decision-making logic into the dataset.

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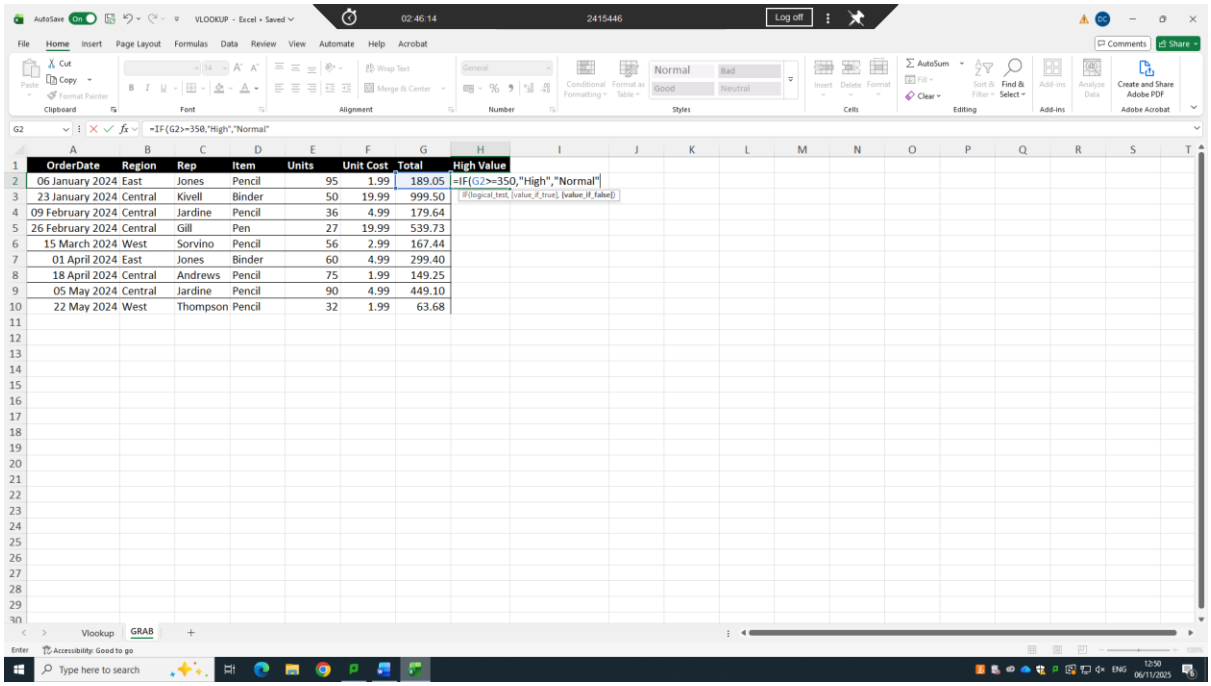

OrderDate	Region	Rep	Item	Units	Unit Cost	Total	High Value
06 January 2024	East	Jones	Pencil	95	1.99	189.05	=IF(G2>=350,"High","Normal"
23 January 2024	Central	Kivell	Binder	50	19.99	999.50	
09 February 2024	Central	Jardine	Pencil	36	4.99	179.64	
26 February 2024	Central	Gill	Pen	27	19.99	539.73	
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Figure 3: IF Statement Before Autofill

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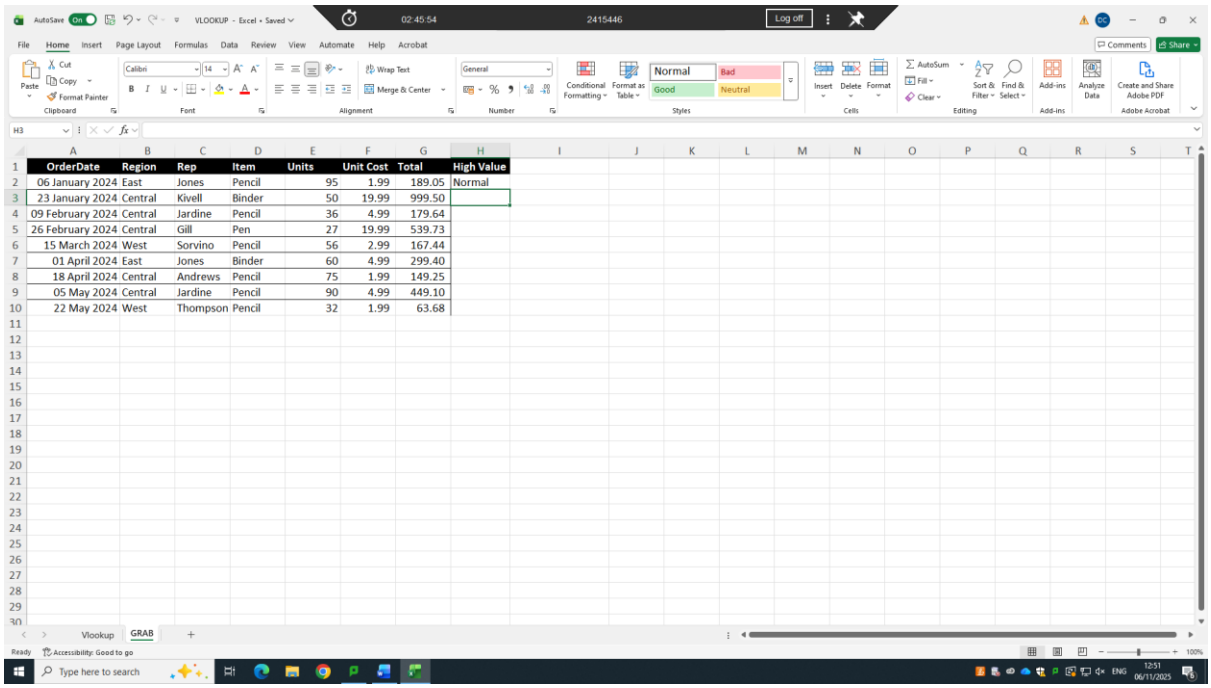
The IF function was initially applied to a single row to verify that the logical condition produced the correct output before extending it to the rest of the dataset.



OrderDate	Region	Rep	Item	Units	Unit Cost	Total	High Value
06 January 2024	East	Jones	Pencil	95	1.99	189.05	Normal
23 January 2024	Central	Kivell	Binder	50	19.99	999.50	
09 February 2024	Central	Jardine	Pencil	36	4.99	179.64	
26 February 2024	Central	Gill	Pen	27	19.99	539.73	
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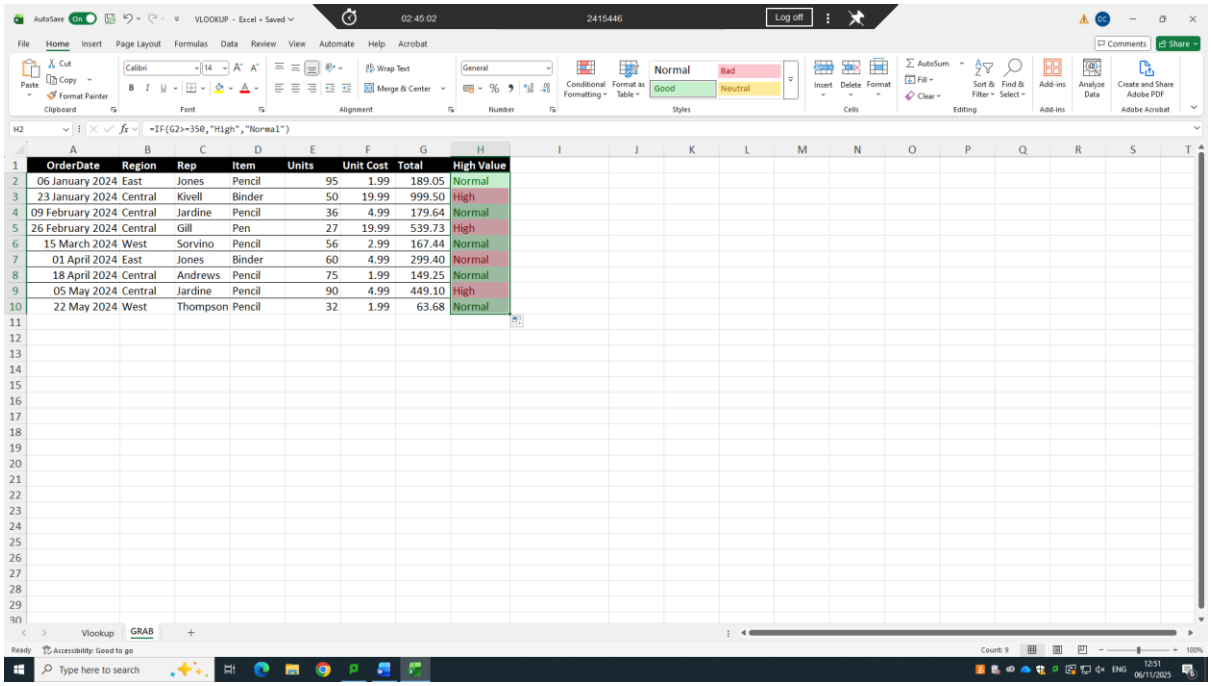
Figure 4 Conditional Formatting

Conditional formatting was applied to visually distinguish high-value orders from normal ones.

This improves readability and allows users to quickly identify significant records without manually scanning numerical values.



OrderDate	Region	Rep	Item	Units	Unit Cost	Total	High Value
06 January 2024	East	Jones	Pencil	95	1.99	189.05	Normal
23 January 2024	Central	Kivell	Binder	50	19.99	999.50	High
09 February 2024	Central	Jardine	Pencil	36	4.99	179.64	Normal
26 February 2024	Central	Gill	Pen	27	19.99	539.73	High
15 March 2024	West	Sorvino	Pencil	56	2.99	167.44	Normal
01 April 2024	East	Jones	Binder	60	4.99	299.40	Normal
18 April 2024	Central	Andrews	Pencil	75	1.99	149.25	Normal
05 May 2024	Central	Jardine	Pencil	90	4.99	449.10	High
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This task demonstrates the practical application of Excel functions to automate calculations, apply conditional logic, and improve data readability. The use of VLOOKUP, IF statements, autofill, and conditional formatting reflects skills commonly required in administrative, data-focused, and entry-level IT roles.